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MAGNETIC POLARITY DURING THE MIDDLE JURASSIC AS RECORDED IN THE SUMMERVILLE AND CURTIS FORMATIONS

MAUREEN B. STEINER *

Institute for Geosciences, The University of Texas at Dallas, Richardson, TX 75080 (U.S.A.)

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Three-quarters of the Summerville Formation and a small portion of the Curtis Formation of central Utah have been sampled at close stratigraphic spacing. The two formations constitute an intertonguing marine–marginal marine, tidal flat sequence representing middle Callovian time. The magnetic polarity sequence observed is dominantly reversed with a few relatively short normal intervals. The age of these formations corresponds to some portion of time within the Jurassic oceanic quiet zone. The fact that normal polarity was not prominent in these formations casts some doubt upon the hypothesis that a long period of normal polarity was the cause of the quiet zone. Furthermore, a literature compilation indicates that normal polarity did not dominate the Jurassic as had been thought. The pole position determined from the Middle Jurassic Summerville Formation slightly revises earlier results and suggests a tiny amount of apparent polar motion relative to North American between Late Triassic and Middle Jurassic time.

1. Introduction

The geomagnetic polarity sequence during Jurassic times is not well known beyond the time of the oldest lineated ocean floor anomalies [1] and the time equivalent Morrison Formation sediments of the western U.S., correlated with these anomalies [2]. Portions of the sea floor older than the oldest magnetic anomalies (the Keathley Sequence in the Atlantic or the Hawaiian-Phoenix lineations in the Pacific) are always magnetically quiet. The lack of anomalies in this portion of the sea floor, the Jurassic quiet zone, is frequently attributed to a lack of reversals of the geomagnetic field and an assumed prevailing normal polarity. Support for this origin is drawn from various summaries of the literature; particularly that of McElhinny and Burek [3], and also those of Perchersky and Khramov [4] and Irving and Pullaiah [5].

Fig. 1 shows a literature compilation of the same type from all published data up to 1976, but an attempt is made to present the data with their assigned ages and age uncertainties, rather than a summary polarity column. Throughout much of the Jurassic, the studies show that both states of polarity have existed; sometimes more determinations are of one polarity or the other are reported for a given time. These data in Fig. 1 are clearly incompatible with the whole of the Jurassic being of single normal polarity sufficient to produce the quiet zone. However, age definition in most of the studies is not sufficiently precise to clearly define a sequence of discrete intervals of polarity of the geomagnetic field. Multiple intervals of alternate polarity in the Jurassic are in agreement with the fact that whenever continuous sampling of Mesozoic time (stratigraphic section) has been done, the detailed pattern that emerges is often similar to the frequently reversing pattern of the Tertiary, i.e., in the Lower Triassic

Contribution No. 335, Institute for Geosciences, University of Texas at Dallas.
* Current address: Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, CA. 91125 (U.S.A.)

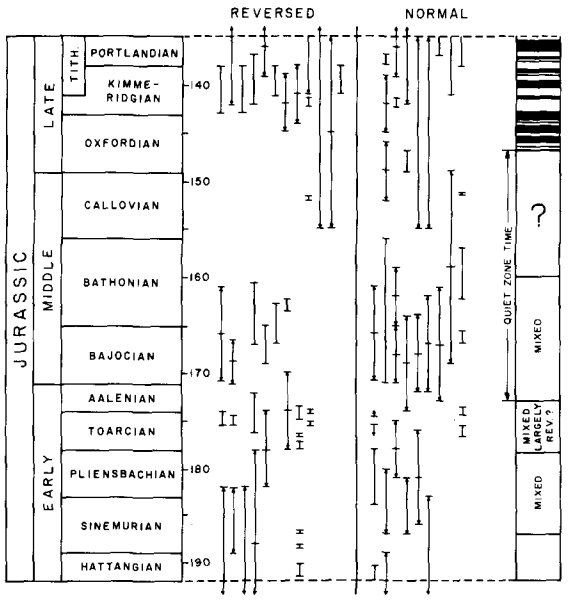


Fig. 1. Summary of published Jurassic paleomagnetic studies. For most of the studies, the age bars indicate the age *uncertainty* of each study, not the age span. Arrows on the age bars denote the plus or minus of radiometrically dated studies. Any study with age uncertainty of greater than 15–20 m.y. is not plotted here. The Late Jurassic mixed interval (upper right) is drawn from the sea-floor anomaly pattern [1] and the Morrison Formation [2]. Time scale is that of Van Hinte [17].

[6,7], the Upper Triassic [8,9], and the Upper Jurassic [10].

Considering the Jurassic from older to younger, Fig. 1 shows that for time earlier than Toarcian, data are few although an equal number of studies report each polarity state. During Toarcian time, there seem to have been frequent reversals as has been noted by others [11,12]; but, at present, reversed polarity has been reported twice as often as normal polarity. In Bajocian to mid-Bathonian time, there are ten reports of normal polarity, and six of reversed. However, most of these data are spot samplings rather than determinations of a continuous polarity record, and thus the dominance of one polarity or the mixture of polarity cannot be determined. For mid-Bathoian to mid-Oxfordian time, there is a paucity of data. For mid-Oxfordian to early Kimmeridgian time, the sea-floor anomaly record (M sequence) and a study of the Morrison Formation [10] show a high-frequency alternating polarity pattern. From Kimmeridgian into Cretaceous time, the sea-floor anomaly record indicates a slightly lower-frequency alternating polarity pattern.

Mid-Bathonian to mid-Oxfordian time corresponds to the time of the younger end of the Jurassic oceanic quiet zone. Extrapolating a constant spreading rate for the oldest part of the M anomaly sequence (M21–M25) in the Atlantic (1.8 cm/yr; R. Larson, K. Klitgord, personal communications, 1977) suggests a duration of 25 m.y. for the quiet zone. The rest of the M anomaly sequence formed at 1 cm/yr [1], and R. Larson (personal communication, 1977) has suggested that the whole sequence may have formed at the 1-cm/yr rate. The quiet zone may then represent as much as 46 m.y.

For the younger third of quiet zone time (assuming 25 m.y.), there is no evidence (Fig. 1) to determine what the polarity state or structure was. For the older two-thirds (to about 173 m.y. in Fig. 1) the data do not warrant the hypothesis of dominantly normal polarity. The present study is aimed at the data gap in the younger third of quiet zone time, and is part of a larger study to develop a detailed polarity history of the magnetic field during the Jurassic. It is intended to sample in detail, continuous sequences of sedimentary rocks formed throughout the 25- or 46-m.y. period of the quiet zone. The studies are aimed at placing another constraint on the many possible origins (see [13–15]) of this quiet zone.

2. Geology and location

The Summerville Formation consists of 120 m of thin-bedded, very evenly-bedded red siltstones and fine sandstones. The sandstones and siltstones are extremely well cemented at this locality. Near the top, some gypsum is present and the uppermost 12–13 m consists of massive gypsum. Overlying the gypsum are red, purple and ocher earthy mudstones of the Morrison Formation. Underlying the Summerville is the Curtis Formation, 65 m of thick bedded green, medium- to fine-grained sandstone. Carbonaceous matter and carbonized wood impressions are abundant in a few horizons within the Curtis. Samples for pollen analysis were collected, but no pollen was found.

The red sandstones and siltstones of the Summer-

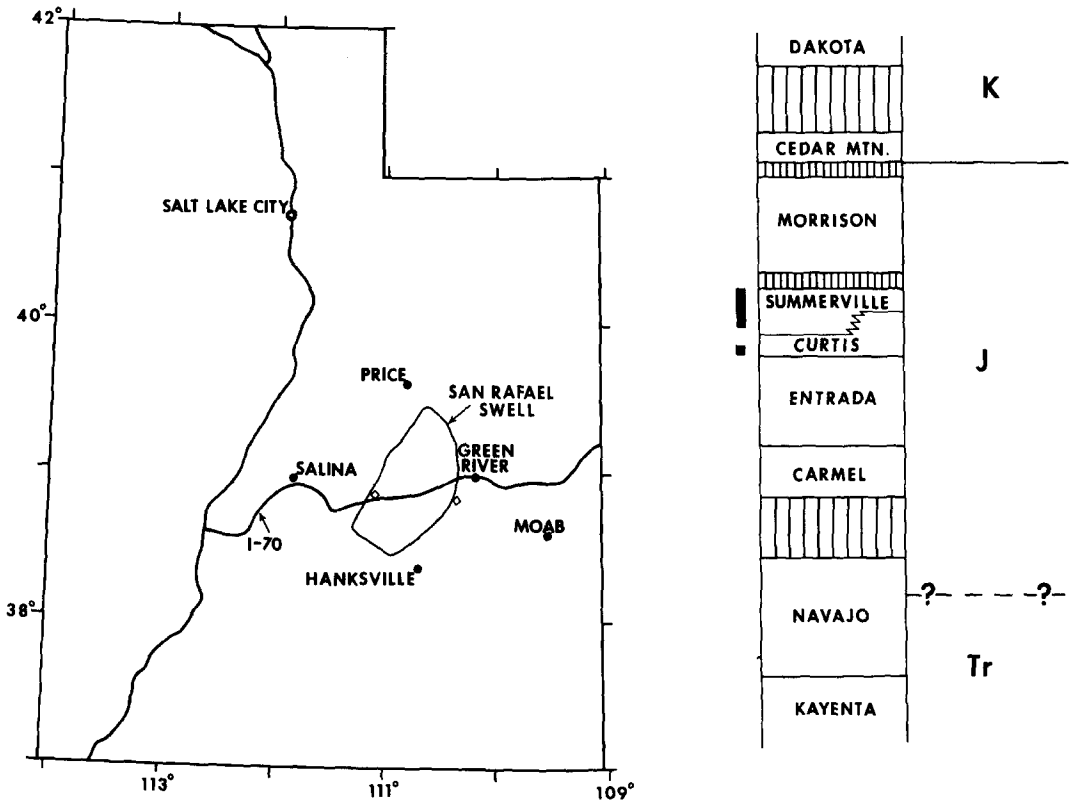


Fig. 2. Location of sections (◇) and stratigraphic relationships.

ville Formation and the green sandstone of the Curtis Formation intertongue, forming a continuous depositional sequence changing from marine sedimentation (Curtis) to marginal marine, tidal flat deposition (Summerville) with time. Neither unit contains age diagnostic fossils, but based on regional stratigraphic correlations relative to fossiliferous units above, below and laterally, the sequence is believed to represent middle Callovian time (R. Imlay, personal communication, 1975). Using the, at best, inexact numbers of published Jurassic time scales, this would be very approximately either 158.5 to 160 m.y. [16], or 151 to 153 m.y. [17].

The Summerville and Curtis Formations were sampled in the San Rafael Swell, east of Green River, Utah (see Fig. 2) from Interstate 70 roadcuts (111.1°E, 38.8°N). The area is part of the flat-lying, little-deformed sedimentary sequence of the Colorado Plateau. The San Rafael Swell is an asymmetrical, domal, anticlinal uplift within these flat-lying sediments, which formed as part of the Laramide uplifts in latest Cretaceous or Paleocene time. Sampling was done on the western flank of the Swell where the strata are almost flat-lying, the dip being 3° to the west. The Summerville was previously sampled by Steiner and Helsley [18] in a very limited exposure on the eastern flank of the Swell.

The collection extends from the basal portion of the overlying Morrison Formation downwards through three-quarters of the Summerville Formation (95.6 m). The lower 24 m of the Summerville and the upper 38 m of the Curtis are not well exposed, but the lower portion of the Curtis Formation crops out in a second I-70 roadcut. Beginning 4 m above the Curtis base, 18.4 m were sampled. Thirty-two samples (6.2 m) were collected from the Morrison Formation, 391 from the Summerville, and 34 from the Curtis sandstones.

3. Sampling and techniques

A portable gasoline-powered engine and diamond coring bit were used to sample the section. Samples were collected at 22- to 45-cm intervals throughout the main stratigraphic sequence. Cores 2.5 cm in diameter were cut into specimens 2.4 cm long. All measurements were made with a cryogenic magnetometer (Superconducting Technology, Inc.). Both alternating field (AF) and thermal demagnetization were used. AF demagnetization utilized a 400-Hz, 3-axis tumbler in a low field area (<200 γ). Thermal demagnetization was done in air, in a non-inductively wound, horizontal furnace. The furnace has an 18-sample capacity with samples spaced 2.5 cm apart and has a cooling chamber with residual field of 5 γ or less. A few samples of Curtis sandstones were also thermally demagnetized in a vacuum of 10^{-5} torr.

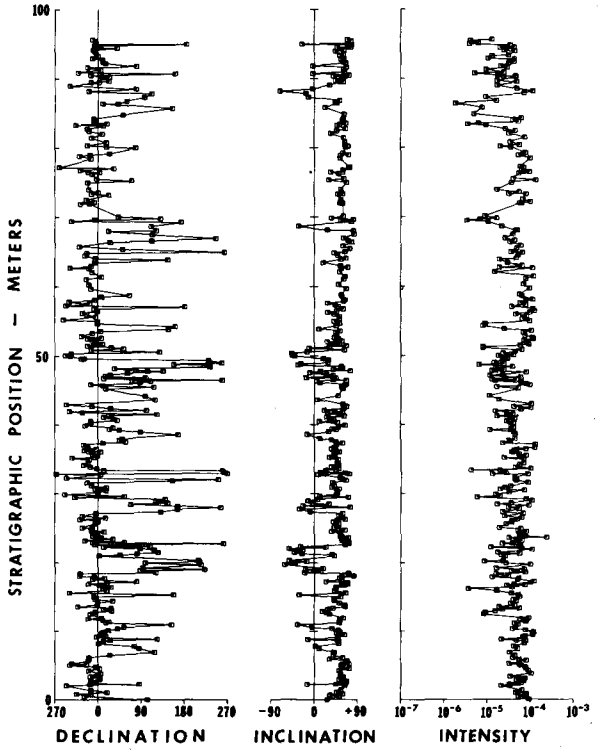


Fig. 3. Stratigraphic plot of NRM directions. Intensity is in gauss.

4. Natural remanent magnetization

The Summerville natural remanent magnetization (NRM) directions mainly lie near the axial dipole field direction (59.1° inclination), but with some scattered between that and an apparently reversed Jurassic direction (Figs. 3 and 4). NRM intensities average about 5–9 $\times 10^{-6}$ G, slightly lower than is usual for red beds.

Fig. 3 shows a plot of all the samples in stratigraphic relationship. Although samples generally have the axial field 0° declinations and 60° inclinations, a few intervals show the reversed affinities of 90° to 180° declinations and negative inclinations. Only one interval (at about 35 m in Fig. 3) has the combination of declinations between 300° and 360° associated with positive inclinations less than 60°, indicating Jurassic normal polarity. These can be seen as distinct from the axial dipole field direction in Fig. 4. In Fig. 4, the lower 230 samples (54 m of Fig. 3) are plotted in stereographic projection. The predominant axial dipole direction is obvious, with some streaked sample directions, a few clearly reversed ones, as well as the normal group (center right plot of the figure). It is clear even from the NRM that the normal interval is distinct from the axial dipole direction and probably represents the Jurassic normal field.

5. Demagnetization results

Both AF and thermal demagnetization were undertaken on four pilot samples, utilizing two specimens per sample. One specimen was demagnetized in stepwise increments of 11, 22, 44, 88, 175, 350, 700 and 1400 Oe, while the other was heated in steps of 400°, 500°, 570°, and 640°C. Neither technique was completely effective in cleaning most of these samples; however, thermal treatment showed a tendency to move sample directions further along great circle paths between NRM directions and an apparent reversed Jurassic direction. Intensities were reduced only to 80% of NRM by AF demagnetization. Thermal treatment reduced intensities to 46%.

After treatment in 1400 Oe, the AF specimens were also thermally demagnetized. In two of four specimens the thermal treatment caused the direc-

tions to move even further along the great circle path established in AF demagnetization. Both of these were samples containing only red pigment, while the other two contained macroscopically visible black grains. (The pigmented samples differed, in that one was heavily pigmented while the other was only pale red.) This suggests that the secondary moment is carried in the pigment and that thermal demagnetization is the more effective way of demagnetizing these red sediments.

As a result of the greater directional movements and intensity changes induced by thermal demagnetization, another 20 pilot samples were subjected to a more detailed thermal demagnetization, using 50–100°C increments from 250° to 660°C. Thereafter all samples were heated at least to 450°, 560°, and 630°C. Large amounts of directional movement were observed in the pilot samples and in the majority of the samples in the collection. Usually 630–660°C were the temperatures at which either a stable end point was reached or was the last point on a great circle path prior to samples becoming too weak to continue treatment. Generally samples continued movement between 630° and 660°, but usually still were not free of secondary magnetization; that is, samples that did not group well, still did not group

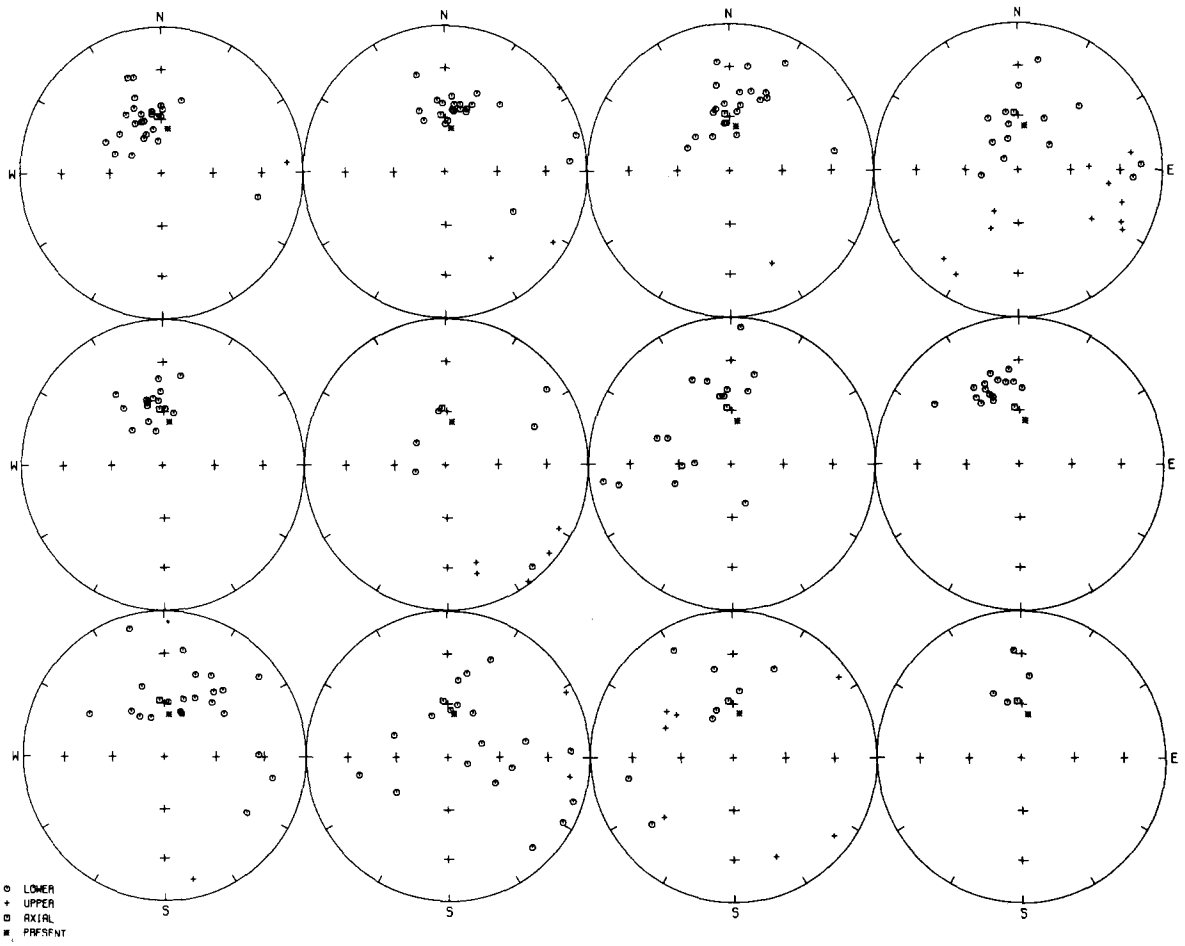


Fig. 4. Equal area projections of NRM directions of the lower 54 m of Fig. 3. The lower samples in Fig. 3 correspond to the top left plot. From left to right and progressing from the top row to the bottom row, the plots depict successive groups of samples from the base of Fig. 3 to 54 m. The selection of groups of samples for each plot was made on the basis of the grouping (or lack) of their demagnetized directions in Fig. 6. "Present" is the observed field at the locality at the present time; "axial" is the computed direction of the geocentric axial dipole. "Upper" and "lower' = upper and lower hemisphere.

well at 660°C, and a number of times the 660°C heating merely produced one further point on the great circle path.

The response to thermal demagnetization is shown in Fig. 5. A few reversed samples and the normal interval at 35 m showed very little directional change in response to thermal demagnetization and had both NRM and demagnetized directions near the previously determined Summerville direction [18] (Fig. 5A). However, the majority of samples show

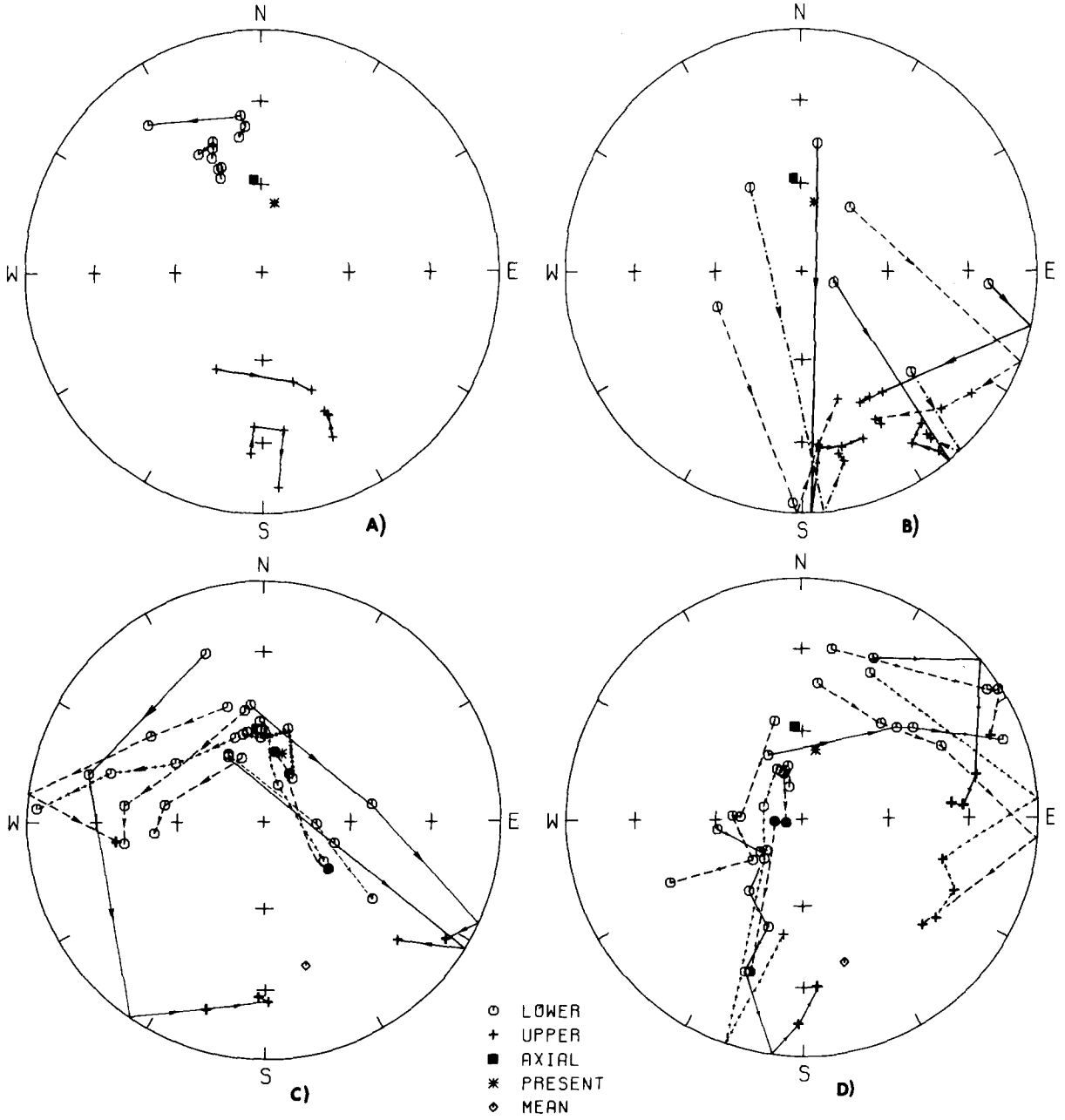


Fig. 5. Response of the Summerville Formation samples to thermal demagnetization. Mean refers to the mean direction that was ultimately derived from this study (see text). Other symbols as in Fig. 4.

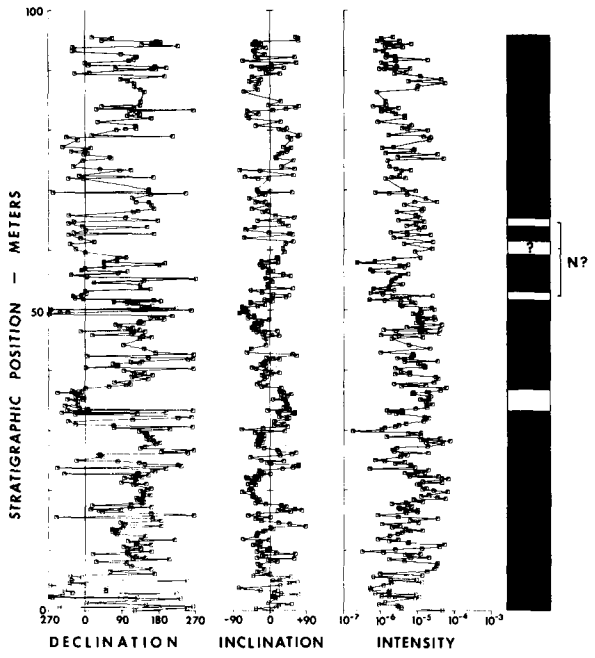


Fig. 6. Thermally demagnetized (630–660°C) directions of the Summerville Formation (intensity in gauss). Black represents reversed polarity.

large amounts of movement away from a direction close to that of the axial dipole field, to a reversed direction. A variety of demagnetization paths were followed as shown in Fig. 5B, C and D. Fig. 5B shows the common unambiguous behavior of motion between the axial dipole field and a reversed direction. However, about 25% of the samples, interspersed in a random fashion throughout the rest of the samples, do not complete the movement to the reversed (or normal) Summerville direction. Examples of this behavior are illustrated in Fig. 5C and D. (Such final sample directions can be seen scattered throughout the plots of Fig. 7.) The polarity of these samples is, therefore less certain, and is assigned by a detailed consideration of each demagnetization path and its similarity to those of samples that went to complete reversed or normal directions. If a sample showed any of the paths illustrated in Fig. 5C or 5D, it was considered to be reversed or probably reversed. The result is then, that most of the 25% appear to be heavily overprinted reversed samples.

The final demagnetized directions (demagnetization to betweeen 630° and 660°) are shown in stratigraphic relationship in Fig. 6. Only three or four intervals have clearly defined polarity. Throughout the sequence, retention of secondary component varies unsystematically, producing alternate sections retaining greater and lesser amounts of secondary magnetization after maximum demagnetization. This alternate retention and the few intervals of relatively well-defined polarity, can be seen in Fig. 7 where the lower 54 m of Fig. 6 are again displayed in stereographic plots. A new plot was used whenever directions either changed polarity, or appeared to increase or decrease in dispersion. The varying response to demagnetization can best be seen by comparing each plot in Fig. 7 with its NRM plot in Fig. 4.

After demagnetization to 630–660°C, most of the section displays either reversed or partially reversed (intermediate between a reversed and axial dipole field) directions. Because of the heavy secondary overprint sustained and partially retained by this rock sequence, samples are not labeled as normal unless there are more than two consecutive ones in a stratigraphic interval, i.e., they represent more than half a meter of section. In this sequence, there is a high probability that a single, apparently normal sample may just be a reversed sample with a hard secondary magnetization in the normal axial dipole field direction.

After a detailed consideration of the demagnetization paths of samples that were not obviously reversed, it appears that, in addition to the obvious normal interval near 35 m, there are three short intervals in the upper part of the section that may be normal (Fig. 6). They are comprised of 3–4 samples each, spanning 0.63–0.73 m. Because they are relatively so short, and because their contained samples do not cluster well, they are labelled N? in Fig. 6. Thus, with the exception of a few short intervals, the entire section seems to be reversed.

6. Uppermost Summerville and Morrison

The uppermost 10 m of the Summerville had been sampled previously (R. Imlay, personal communication, 1975; locality 1). The uppermost part of the section was sampled again just beneath, and including, the overlying basal Morrison Formation at this locality (locality 2). In addition, it was also sampled

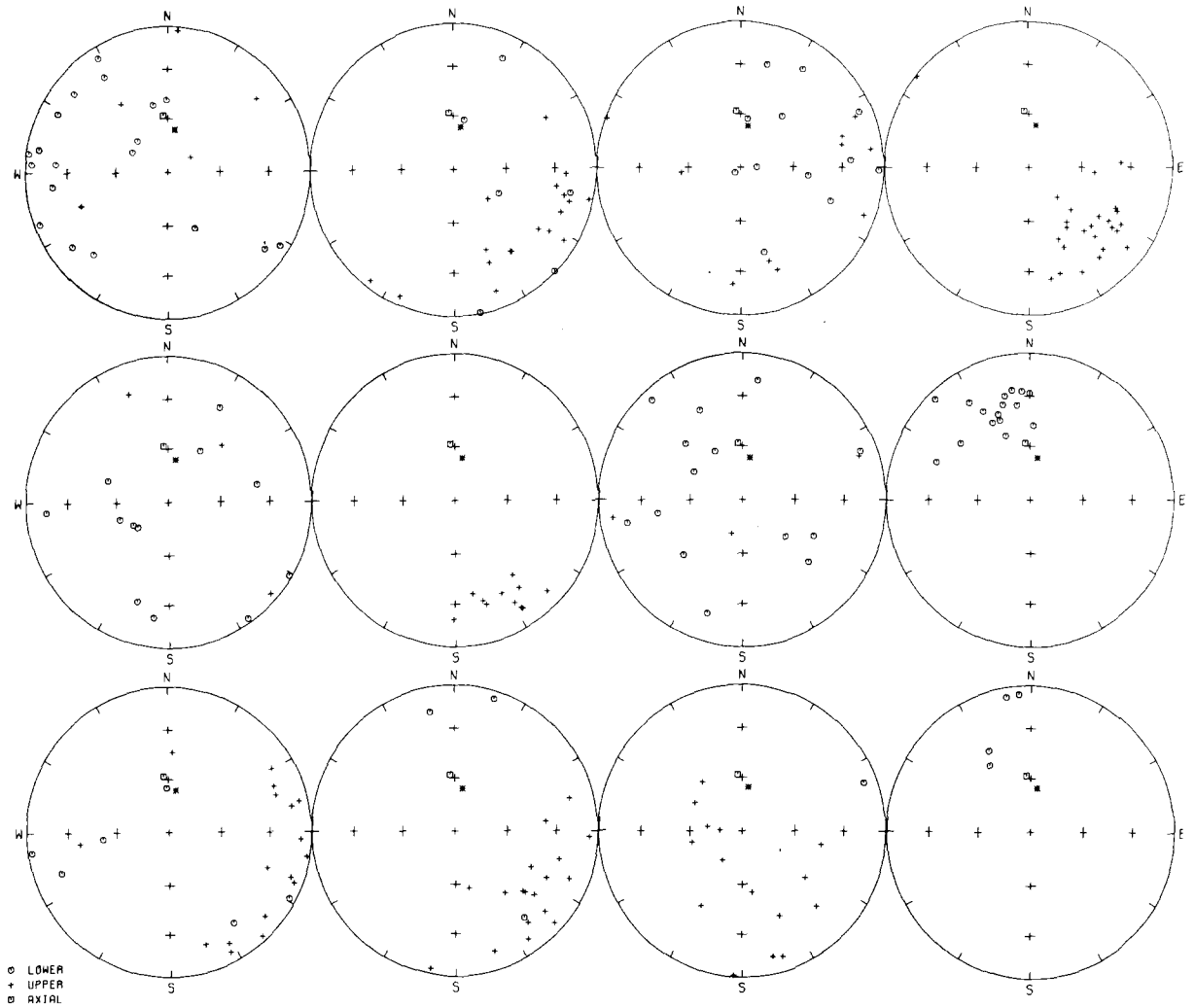


Fig. 7. The lower 54 m of Fig. 6 in stereographic projections (see text). The samples in each plot are the demagnetized directions of the samples in corresponding plots in Fig. 4. Symbols as in Fig. 4.

at a site 26.7 km to the southwest of locality 2 (locality 3), again from the base of the Morrison downward. At the top of the roadcut of locality 2, the five exposed meters of the basal Morrison, which were also sampled, gave directions that are entirely that of the axial field direction. Most did not change direction at all upon thermal demagnetization to 630°C; a few scattered randomly.

A noticeable difference in demagnetization behavior is observed in the uppermost beds of the Summerville at the three localities. At locality 3, the 5 m beneath the Morrison Formation have demagnetization paths quite similar to the top beds in the interval 7–10 m beneath the gypsum at locality 1, and are rather distinct from the demagnetization paths of the bulk of the formation, in that they trend due south and somewhat southwest from an axial field direction to a reversed Jurassic direction. After cleaning they yield a direction more westerly than the Summerville direction obtained from the bulk of the formation. A few of the uppermost samples at locality 2 beneath the 13 m of gypsum (about a meter of section) also exhibit behavior similar to localities 1 and 3.

This shift in direction is similar to ones in the Lower Triassic Moenkopi and Upper Triassic Kayenta Formations, and might perhaps suggest a continuation of the same Triassic field behavior into Middle Jurassic time. More obviously though, the data support the stratigraphic relationships proposed by Dr. Fred Peterson of the U.S. Geological Survey (personal communications, 1973, 1974). Localities 1 and 3 are farther south than locality 2. Their peculiar directions are recorded quite close to the top of the formation just beneath the overlying Morrison Formation, while at locality 2 there is a large thickness of gypsum with some interbedded siltstones, between the few peculiar sample directions and the top of the

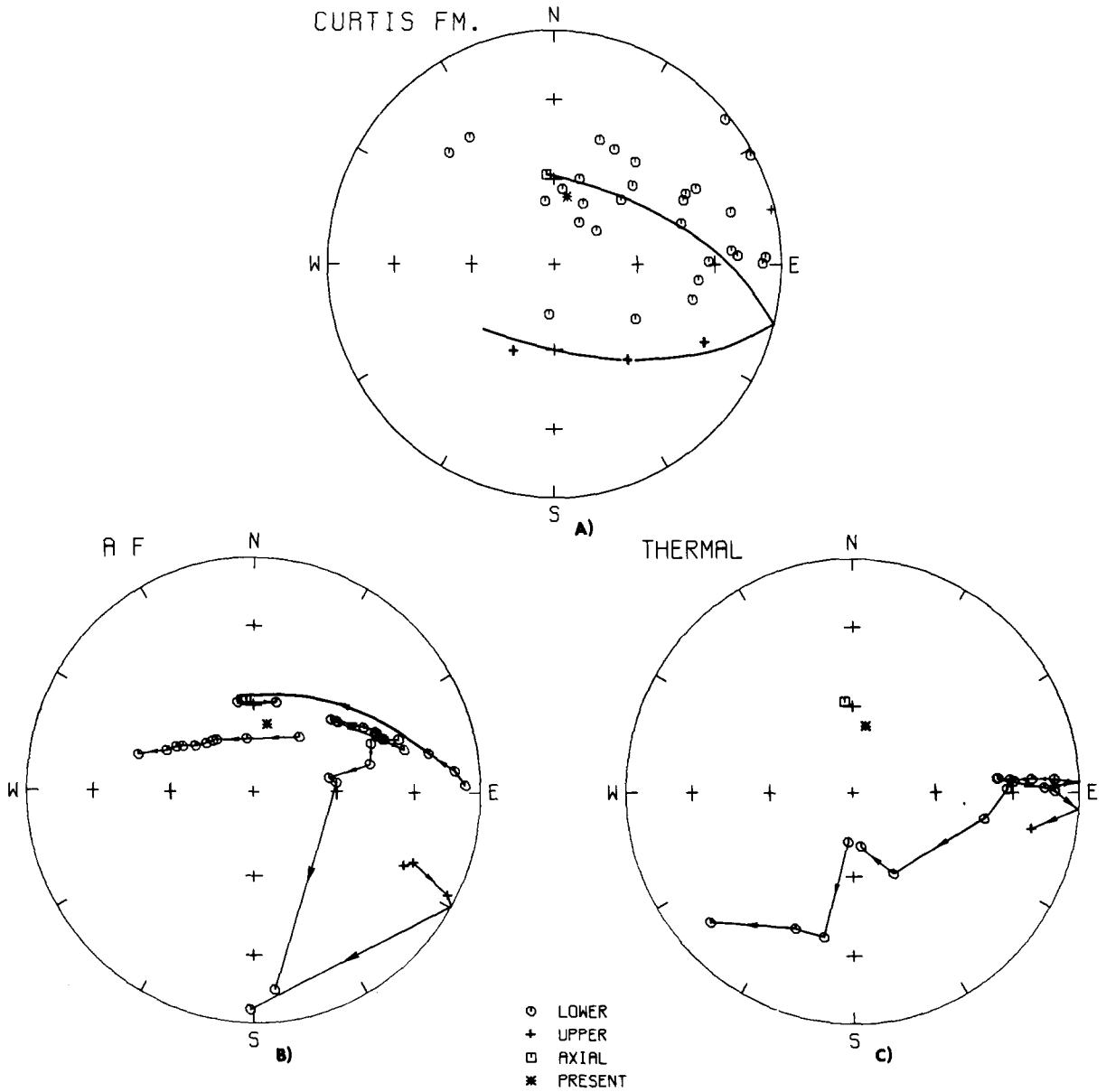


Fig. 8. Curtis Formation: (A) NRM directions, (B) response to AF demagnetization, (C) response to thermal demagnetization. Symbols as in Fig. 4.

formation and overlying Morrison. If these different directions can be considered to be time equivalent (and thus a time line), this supports the idea that the locality 2 area was near the center of the Summerville depositional basin and became an evaporating pan as the Summerville sea dried up.

7. Curtis Formation

The Curtis Formation is magnetically perplexing. Because it is quite friable and pale green in color, it was sampled in a preliminary fashion at large spacings of 1 m, and sometimes 2–3 m. The 32 samples give a rather interesting NRM plot (Fig. 8A). The directions seem to form a streak between the axial or present field direction and a reversed Jurassic direction. Several samples have reversed, negative inclination NRM directions and two have apparent normal directions.

However, cleaning techniques produce undecipherable results. Both AF and thermal demagnetization were used, heating some in vacuum (10^{-5} torr) and some in air. All samples were AF demagnetized in steps to 88 or 175 Oe. Nine were heated to 100°, 200°, and 300°C. Several patterns of movement are exhibited on both types of demagnetization, however, only a few samples exhibit paths suggesting normal or reversed magnetization overprinted by a normal axial field magnetization as was the common behavior in the Summerville Formation. Fig. 8B and C show examples of behavior in response to these demagnetization treatments. In response to 175 Oe peak field demagnetization intensities decreased to 20–50% of NRM, commonly 30%. On heating intensities decreased also to about 30% of NRM. All that can be said from the demagnetization behavior is that there may be a softer primary (reversed?) magnetization which has been overprinted by a harder secondary magnetization, having both higher blocking tem peratures and coercive force. It is possible that the peculiar direction paths and the tendency toward randomness at higher fields results from a combination of a very low concentration of magnetite, such that the total intensity contribution decays to negligible values in low fields, and a low concentration of hematite which resists the lower fields and temperatures but is magnetized in more than one (Cretaceous and Tertiary) overprinting field. Petrographic examination of a few polished sections confirmed the presence of a tiny amount of hematite.

8. Mean Summerville direction

After thermal treatment, there is still a high dispersion to the directions of the Summerville samples. This makes it difficult to tell precisely what the exact direction of magnetization is. It is desirable to know this direction as very little is known of the magnetic direction between Late Triassic time and Late Jurassic time [10,18,19]. It seems that the magnetic direction remained similar from Late Triassic to Late Jurassic (including Summerville) time ([18], see also Johnson [12]). It is also desirable to compare the direction from this locality with that of the limited first study [18]. Thus, various degrees of selection of the data have been preformed in an attempt to get a better estimate of the real Summerville direction. The selection has centered around the criteria of Helsley ([20], also Scott [21]). i.e., the farther away from the estimated mean direction the NRM direction of a sample is, the greater the likelihood that the sample will retain hard secondary magnetization after demagnetization. While this selection criterion assumes the same stability ranges in the secondary moment from sample to sample regardless of the quantity of secondary moment added, this does not always seem to be the case in the Summerville samples. It is sometimes observed that samples very far removed from the estimated mean direction demagnetize to directions very close to the mean while stratigraphically adjacent samples with NRMs closer to the mean do not clean well at all.

The results of the selection technique can be seen in Table 1. The selection technique requires an estimated mean (end of the streaked distribution between primary and secondary directions). The allowable range of directions that could be considered as primary Jurassic magnetization can be seen in Fig. 9A. (Only formations from the Colorado Plateau are considered because of an apparent discrepancy between Colorado Plateau and eastern U.S. Late Triassic poles; see [8,9,19].) The range is between Late Triassic and Late Jurassic directions, with Cretaceous also shown. The mean was first estimated by

TABLE 1

Directions of magnetization and pole position of the Summerville Formation

	Declination (°)	Inclination (°)	*N*	*R*	*K*	α_{95}	Long. °E	Lat. °N	*dp*	*dm*
Cleanest reversed	163.6	−36.3	5.0	4.98	165.59	5.96	110.7	66.6	4.04	6.94
Cleanest normal	344.5	37.9	10.0	9.83	52.57	6.72	110.6	68.0	4.69	7.94
Cleanest N + R *	*164.2*	*−37.4*	*15.0*	*14.80*	*71.07*	*4.57*	*110.6*	*67.5*	*3.16*	*5.37*
<37° away	154.2	−38.4	29.0	27.76	22.52	5.77	128.5	62.0	4.06	6.84
<90° away	154.1	−38.1	33.0	31.70	24.60	5.14	128.3	61.8	3.60	6.90
<120° away	153.1	−35.5	39.0	37.17	20.76	5.15	127.0	59.9	3.45	5.96
1972 Summerville cleanest N + R	166.4	−33.3	8.0	7.72	24.97	11.30	104.7	66.9	7.31	12.86
Sum of 1972 and 1977 localities cleanest N + R	164.2	−36.4	23.0	22.57	51.46	4.26	109.8	67.0	2.90	4.97

* The pole considered most representative of the direction of the Summerville Formation. "'Cleanest" in this study = <25° away; <40° away for 1972 study.)
N = number of samples.

taking a mean of all samples, normal and reversed, whose NRM direction was in the vicinity of the expected Jurassic direction and which did not move upon demagnetization. The NRM directions of these individual directions are not more than 25° removed from their calculated (demagnetized) mean. As mentioned above, very hard secondary magnetization is retained at various points throughout the section. Thus, any apparently stable sample (unchanging upon demagnetization) with NRM directions substantially removed from the group of Triassic through Cretaceous directions, was not included in the data group of the initial mean calculation. Separate means of the normal and the reversed directions of this group give identical north pole positions (see Table 1). This is strong evidence that this mean is a good estimate of the Summerville direction.

After this first estimate, mean directions were calculated for data whose NRM directions were less than 37°, 90°, and 120° away from this estimated mean. These numbers were suggested by the distribution of the NRM directions of the data points.

As can be seen from Table 1, the mean directions calculated from the addition of samples with increasingly greater amounts of initial secondary moment are *all* significantly shifted from the original estimate. This fact is also observed in simply examining individual sample demagnetization paths, i.e., all samples containing secondary magnetization sufficient to cause their NRM directions to be greater than approximately 25° away from the mean of the least overprinted samples, often do not move on demagnetization quite to the direction of those samples that contained little secondary magnetization. Thus it would seem that sample directions that had any appreciable secondary magnetization are suspect, and that the best estimate that can be made of the mean direction and pole position is that from the least remagnetized samples (<25° away in Table 1). This is especially true in light of the antipodal nature of the normal and reversed samples.

The same selection process applied here has now been applied to the older Summerville results (of Steiner and Helsley [18]). It should be mentioned here that the poles quoted in the original paper have a latitude error of the site of 1°. A pole for the older data is listed in Table 1 as calculated from this selection process and with the correct latitude.

With the amount of hard overprinting present in the Summerville, any mean or pole position is approximate at best. Nevertheless, I would argue that the combined pole from the normal and reversed samples of this study may be a reasonable approximation of the pole at this time. This argument stems from the perfect antipodal nature of the means of widely stratigraphically separated normal and reversed sample directions, and the very close agreement of the two site poles, 110.6°E, 67.5°N and 104.7°E, 66.2°N

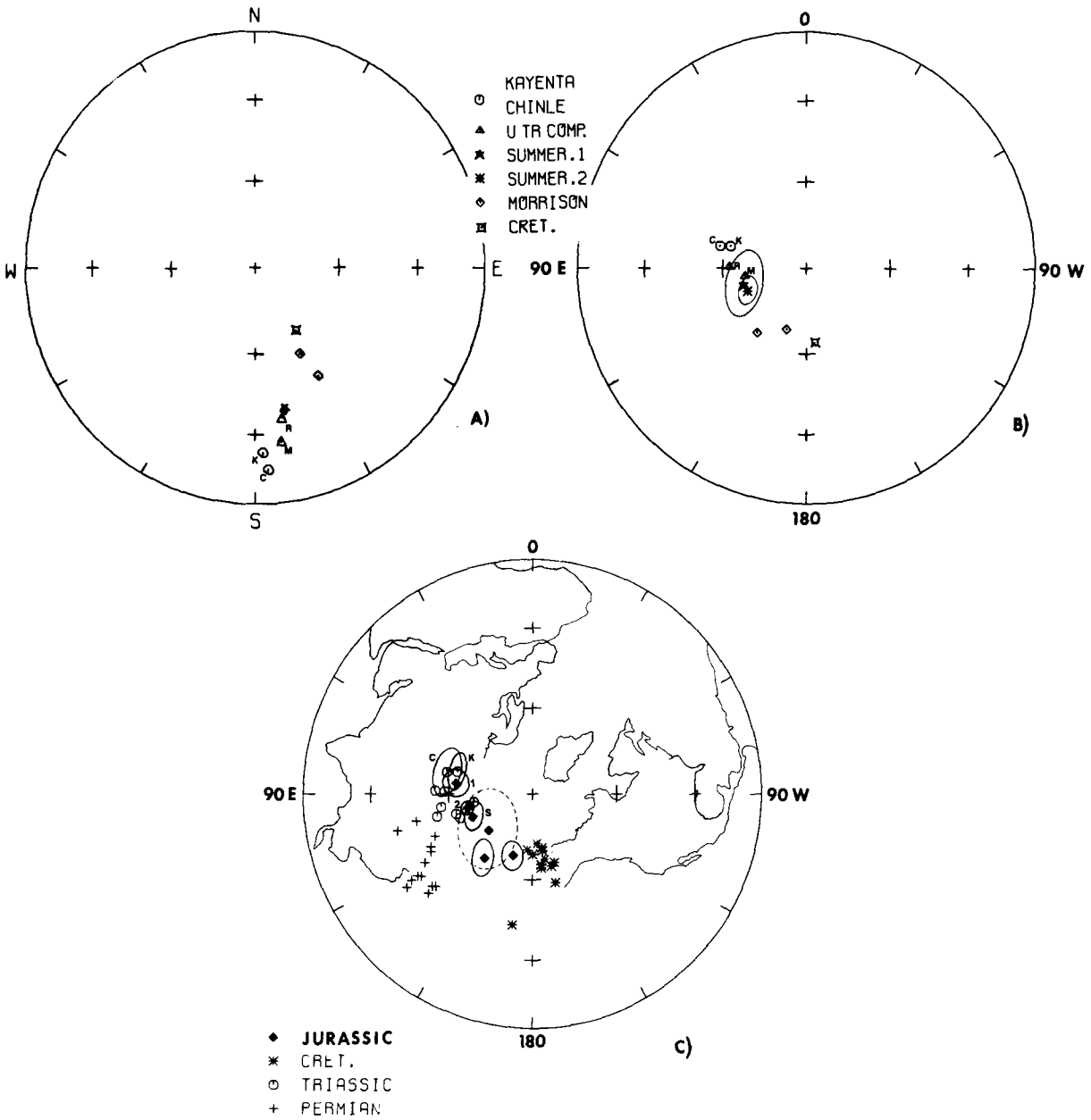


Fig. 9. A. Directions as viewed at this locality for times older and younger than Summerville time. K = Kayenta, C = Chinle Formations [9,8] (Triassic formations); U Tr Comp. = Upper Triassic composite computations (R = Reeve [8]; M = McElhinny [22]); Morrison = Steiner and Helsley's [10] poles; Summer 1 = Steiner and Helsley's [18] older Summerville Formation pole; Cret. = McElhinny's [22] Cretaceous pole for North America. "Present" and "axial" are as in Fig. 4.

B. Pole positions from Late Triassic to Cretaceous time, showing a comparison of the pole position and oval of confidence of this study (2) and that of the previous study (1) of the Summerville Formation. Symbols as in A.

C. Fig. 2 of my earlier discussion of apparent polar motion [19] with the substitution of the new pole position and oval of confidence of the Summerville Formation. The oval of confidence has been added to the Kayenta (K) and Chinle (C) pole positions. (Pole 4 of the old figure has been revised and is included in the poles of Smith and Noltimier [23].) Poles 1 and 2 are those of Smith and Noltimier [23].

(see Fig. 9B). For a statistically better (more samples) pole one could average the two locality directions together. This is done in Table 1. However, the NRM directions of those stable normal and reversed samples of this locality are all within 25° of their demagnetized mean, while those from the 1972 locality have higher dispersion (dp = 7.3, dm = 12.7) and only had NRM directions within 40° of their mean (see Fig. 9B). Thus, the mean of the normal and reversed samples from only this locality is probably a better representation of the Summerville direction and pole position (110.6°E, 67.5°N; dp = 3.2, dm = 5.4), and may be a good representation of the geomagnetic pole at this time.

The pole position of the earlier study suggested that apparent polar wander relative to North America was minimal from Late Triassic through Middle Jurassic time [21]. The new pole position from this study and the newly computed position from the older study suggest that a very small amount of apparent polar wander relative to North America may have occurred between Late Triassic and Middle Jurassic time. In Fig. 9B pole positions and ovals of confidence of the two Summerville studies are plotted with the Late Triassic Chinle [8] and Kayenta [9] Formations, as well as two published estimates of the Late Triassic pole position for North America. Relative to McElhinny's estimate of the Late Triassic pole, it would appear that no apparent polar wander had occurred between Late Triassic and Middle Jurassic (Summerville) time. However, that Late Triassic pole includes many determinations made on eastern U.S. Late Triassic units. As previously discussed [19], eastern U.S. rocks often given a more northerly pole position which has not been observed in Colorado Plateau formations. If a direct comparison of the early Late Triassic Chinle Formation and the late Late Triassic Kayenta Formation poles with the Summerville pole from this study is made, about 10° of apparent polar motion is suggested, probably during early Jurassic time.

This conclusion is in striking agreement (Fig. 9C) with an even more recent study [23] in which two groups of pole positions are found to exist, one of Late Triassic–Early Jurassic age, which is close to the Kayenta–Chinle pole positions, and the other of Early to Middle Jurassic age, which is very similar to the late Middle Jurassic Summerville pole position. The present study and the more recent study [23] both suggest a small amount of apparent polar motion relative to North America during the Early to Middle Jurassic. This motion appears to be the precursor to the later rapid apparent polar motion [19, 10] related to the large-scale opening of the Atlantic, and forms a continuous path with that later motion (Fig. 9C).

African data are not in disagreement with, and hint at a very similar apparent polar motion, but it is not presently as clearly resolved. If the African data do show similar motion, they are stationary thereafter, in agreement with my previous postulation that the opening of the Atlantic was due to asymmetric spreading primarily involving motion of the North American plate, as reflected in the motion of its apparent polar wander curve versus the stationary African poles [19]. If, as suggested here, both continents show the same Late Triassic–Early Jurassic apparent polar motion, a preliminary symmetrical spreading of the Atlantic is implied prior to the asymmetric motion of the North American plate.

9. Discussion and conclusion

The Summerville Formation has revealed both an unusual polarity sequence and a pole position that produces a slightly different interpretation of the North American polar wander curve. The pole position, although drawn from a small number of samples, suggests, in agreement with a more recent study [23], a small amount of apparent polar wander possibly implying a small amount of initial symmetrical spreading of the Atlantic during Early to Middle Jurassic time.

In polarity, the majority of the Summerville directions seem to be reversed. This is the simplest interpretation of these heavily overprinted data, although other interpretations are perhaps possible. There is at least one, relatively short, normal interval in the present study, and possibly several more even shorter ones (Fig. 6). The 2 m of normal polarity observed at the top of the section previously studied [18, fig. 11] is not apparent here and probably is equivalent to time preserved here only as massive gypsum deposits. Alternatively, a short normal might exist in some of the heavily overprinted directions in the

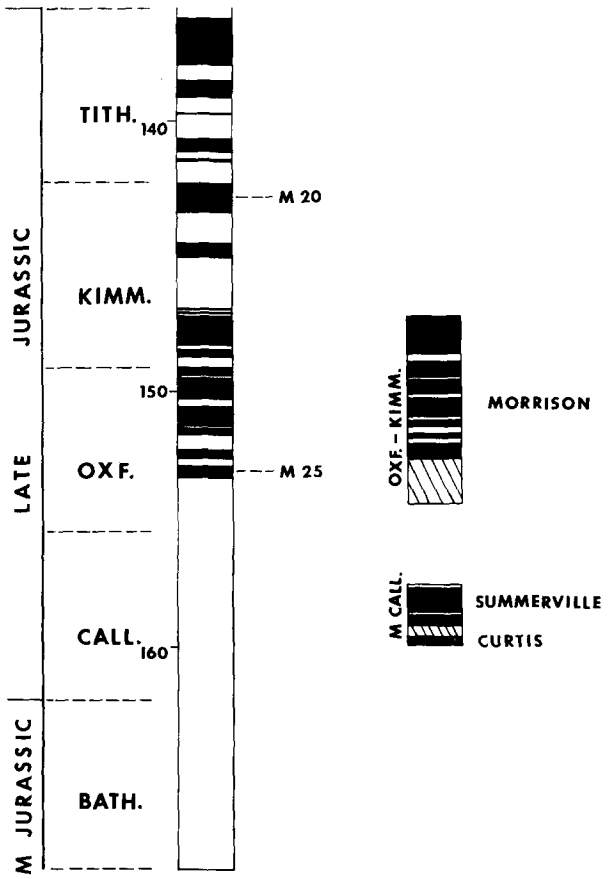


Fig. 10. Summary of Late to Middle Jurassic reversal knowledge. Left column is from Larson and Hilde [1] with their assigned ages. Right column is stratigraphic studies of sedimentary rocks with assigned ages. (Van Hinte's [17] time scale is preferred for the Jurassic as it is a composite of biostratigraphic and radiometric information while the London Geological Society time scale [16], used by Larson and Hilde [1] and reproduced here, is based only on radiometric information and requires large-scale extrapolations. Nevertheless, it is retained here since it is the time scale that has been used to assign ages to the sea-floor anomalies and quiet zone. However, these ages may not be accurate, and the sea-floor ages may need reevaluation.) Diagonal lines indicate unsampled intervals of section. Normal is white, reversed is black for emphasis.

upper portion of this section. Nevertheless, the Summerville polarity sequence has at least one additional normal interval at or near its top (Fig. 10). The Curtis Formation data, while ambiguous, are more suggestive of reversed than normal polarity. The Summerville and Curtis directions suggest that the middle Callovian portion of the magnetic record is predominantly reversed. Together with the Morrison Formation results, also more reversed than normal [2], these data further suggest that the late Middle and early Late Jurassic may have had a preponderance of reversed polarity. Projected against the sea-floor spreading anomaly pattern in Fig. 10, these stratigraphic sequences depict the polarity data so far known for the Middle and Late Jurassic.

This reversed polarity occurring in time that is equivalent to time during the formation of the Jurassic magnetic quiet zone disagrees with the general view that the quiet zone represents a long period of normal polarity. Neither these data (Fig. 10) nor detailed consideration of the published literature (Fig. 1) support that view. Although Summerville and Curtis deposition probably only represent a relatively short time (several million years based on consideration of ages of fossiliferous formations stratigraphically above and below), studies in progress [24] seem to substantiate a preponderance of reversed polarity. At present it seems that during the time in which at least the younger half of the Jurassic oceanic quiet zone was formed, reversed polarity of the geomagnetic field may have been predominant.

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